Designing the Disengagement Experience in Short-Form Video Use: A Diving-Metaphor Prototype

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Abstract. Short-form video platforms have been extensively optimized for engagement, yet much less attention has been paid to how users leave them. Drawing on prior work that frames disengagement as an experience that can itself be designed, this paper explores whether short-form video use can support a more noticeable and agentic form of disengagement through metaphorical gameful design. Thus, we developed a prototype namely "Scroll Dive" that reframes browsing as a process of diving underwater: users enter the feed by "diving," remain within it as oxygen gradually depletes, and exit by "surfacing." Rather than relying on direct friction or interruption alone, the design aims to embed stopping within an internally coherent experiential frame. We conducted a preliminary study comparing the prototype with a frictional design condition. The findings suggest that the prototype was more effective in making disengagement feel noticeable and agentic, with stronger effects on perceived choice and arousal. The paper contributes an early design exploration of positive disengagement in short-form video use and outlines directions for future research on exit-oriented interaction design.

Keywords: Short-form Video, Design for Disengagement, Gameful Design.

1 Introduction

Short-form video (SFV) platforms have become one of the most popular forms of everyday entertainment [1]. Globally, 87.5% of adults report watching short videos at least once a week, spending an average of 57 minutes per day. This popularity is driven not only by SFV content, but also by the frictionless design of the app interface [2]. Features such as infinite scroll, autoplay, and swipe-based navigation make these platforms highly effective at sustaining continuous attention [3]. Users may continue watching beyond their original intention, not necessarily consciously, but because the interface offers few natural cues for disengagement [4]. SFV use therefore raises a critical design problem: not only how attention is captured, but also how the experience of leaving might be shaped.

Despite this, far less attention has been given to how users disengage from these platforms. Research on games suggests that disengagement should not be understood

merely as interruption or failure, but as an experience that can itself be constructively designed [5].

This paper explores the possibility of designing for more constructive disengagement in SFV use through a prototype structured around a gameful diving metaphor. We offer a preliminary investigation into how disengagement support might be designed for SFV use. Its contribution is not to claim long-term prevention of overuse, but to provide an initial test of whether disengagement can be supported in a way that is more gameful, perceptual and noticeable, while fostering greater sense of closure and autonomy.

2 Related works

2.1 SFV as hard-to-exit systems

Short-video platforms are often understood as engagement-optimized systems designed to maximize users' watch time [6]. Research suggests that features such as infinite scroll, autoplay, and algorithmically feeds reduce decision-making demands and support prolonged interaction. Alruwaili identifies "scroll immersion" as a form of habitual and unintentional engagement and shows that it is associated with attention difficulties, working-memory disruption, and cognitive fatigue [7].

One reason these systems are difficult to exit is that they enable frictionless continuation while weakening stopping cues [8]. Infinite scroll supports extended sessions by reducing the mental and physical effort required to continue, while autoplay further compresses the time available for reflection, breaks, or deliberate choice [8, 9]. In streaming research, users reported that rapid autoplay countdowns pressured them to continue before fully processing what they had just watched. Similarly, in social media studies, participants described default feeds as encouraging continued scrolling unless explicit stop criteria were reintroduced [9, 10]. Related work on infinite-scroll interfaces likewise links these designs to mindless scrolling, reduced awareness, and a sense that one's original goals for the session were not ultimately fulfilled [8].

Research further suggests that hard-to-exit environments also weaken the cognitive conditions required to remember and enact the intention to stop. Chiossi et al. showed that SFV feeds significantly impaired prospective memory relative to Twitter, YouTube, and no-media conditions. This finding indicates that the combination of brief videos and rapid context switching may disrupt users' ability to recall and act upon prior intentions, including the intention to disengage [11].

2.2 Design for disengagement

Recent work has framed disengagement as a necessary phase within the dynamics of engagement rather than merely a sign of breakdown [12, 13]. Alexandrovsky et al. argue that disengagement from games can itself be designed in ways that support self-determined exit. Their study shows that positive disengagement is associated with satisfaction and closure, while agency and a sense of control are central to how players experience leaving play [5]. From this perspective, leaving is not simply the termination

of engagement, but an experiential phase whose quality can be intentionally shaped and evaluated.

Related arguments have also emerged in broader media studies. Baumgartner and Kühne note that theories of media use have historically been much better at explaining selection and engagement than disengagement, despite the fact that media use unfolds as a temporal process with both a beginning and an end [14]. They further suggest that stopping is not determined solely by external interruption but can also arise from the changing qualities of the experience itself over the course of use [14]. Ernst and Schnauber in recent study identified internal, environmental, and media-context cues that prompt users to stop on TikTok and Instagram, showing that disengagement may take multiple forms rather than converge on a single endpoint [15].

Although current research increasingly recognizes disengagement as a multidimensional experience, it remains unclear how SFV interfaces might actively support a more constructive exit experience. This paper therefore explores whether SFV disengagement can be made more noticeable and agentic through a metaphorical gameful design, with attention to closure, perceived choice, valence, and related affective appraisal.

3 Prototype Design

3.1 Design intention

The prototype was designed as a disengagement-oriented interface for SFV use. Its main purpose is to help users end a viewing session at a coherent stopping point, in a way that feels self-directed and positively appraised after exit. At the same time, the design also introduces a persuasive limit on continued viewing: the diver's remaining oxygen functions as a representation of recommended viewing time, making session duration visible and giving users an interpretable point at which disengagement is encouraged.

To do so, the prototype adopts a diving metaphor as a gameful disengagement mechanism. By mapping short-video entry, continuation, and exit onto a bounded dive process, the design frames use as a finite session rather than an open-ended stream. Within this structure, the oxygen system serves not only as thematic feedback but also as a soft regulatory cue, signaling the session's limited duration and gently encouraging users to stop without forcing termination. In this way, the prototype is intended to make disengagement more intentional and pleasant, with improvements in the disengagement experience primarily reflected in greater perceived closure, control and pleasure.

A demonstration of the Scroll Dive is available for viewing at the following YouTube link. ***https://youtube.com/shorts/TRNsIYs9DTI?feature=share***

3.2 Interaction flow

The prototype structures short-video use as a bounded dive process. Its interaction flow is organized as follows (see **Error! Reference source not found.** and the Youtube link above):

1) **Pre-dive state**
   A 3D scene in which a diver rests calmly at the water surface. The UI displays Total Underwater Time and Oxygen Remaining. The former indicates the user's total SFV viewing time, while the latter represents the recommended maximum time for a single viewing session. The user can swipe down anytime to begin diving.
2) **Diving & viewing state**
   SFV content will then occupy most of the screen, while the diving scene will shrink into a small floating window. As viewing continues, the diver's oxygen gradually decreases. This depletion functions as a visible progression cue, giving the session a sense of duration and finiteness.
3) **Approaching maximum time cues**
   When oxygen nears a threshold, warning signals appear through animation, lighting changes, and visual effects indicate that the session is approaching to suggested maximum time.
4) **Ascent**
   At any moment, the user may swipe up to trigger ascent. The diver then plays an ascent animation, while the diving scene expands from a small floating window back to full screen.
5) **Return to the surface**
   After ascent, the interface returns to the pre-dive screen. The UI encourages users to take a few seconds of deep breaths before starting another viewing session, while the Oxygen Remaining gradually recovers to its full level.

3.3 Disengagement supportive mechanism

The design seeks to support three factors closely related to exit experience.

- **Closure.**
  Closure is supported by the overall dive structure, which gives viewing a clear beginning, progression, and end. The pre-dive surface state separates readiness from immediate consumption, oxygen depletion makes the session feel finite, and the ascent-return sequence turns leaving into a complete transition.
- **Perceived Control.**
  It is supported through character control, a key experiential quality often emphasized in games. In the prototype, users do not merely consume content; they control the diver's movement by initiating descent and performing ascent. This mapping of browsing onto character action makes disengagement feel enacted by the user rather than externally imposed. Although the oxygen system introduces a persuasive limit, it does not force exit, allowing agency to remain with the user.
- **Affective appraisal.**
  It is supported by turning disengagement into a more aesthetically and experientially rewarding process rather than an abrupt interruption. The diving metaphor, oxygen feedback, and ascent sequence work together to make leaving feel less resistant than simply stopping an endless feed.

4 Preliminary User Study

We present a preliminary study that examines whether the prototype Scroll Dive can support the three intended exit-quality dimensions identified above, namely closure, perceived control, and post-exit appraisal. As an initial investigation, the study is necessarily limited in its overall methodological scope, including its study design, sample size, and control conditions.

4.1 Research Questions

Accordingly, the preliminary study addresses the following research question:

RQ: Does Scroll Dive support a positive exit experience in short-form video use, in terms of closure, perceived control/agency, and post-exit appraisal?

4.2 Study design

Participants. This preliminary study employed a between-subjects design with two conditions: a Scroll Dive condition and a control condition. The study included 25 participants in total, with 13 participants assigned to the Scroll Dive condition, and 12 assigned to the control group.

Conditions. Two conditions were used in the study: the Scroll Dive condition and a frictional control condition. The control condition was constructed by removing the core diving metaphor and replacing its game-like interface expressions with neutral, non-metaphorical ones. In this condition, disengagement support was conveyed primarily through frictional cues. The video pool was constructed from 50 animal-themed short videos selected from highly popular TikTok content because they are less dependent on topical preferences than categories such as politics, beauty, or gaming. In both conditions, the Recommended Maximum Watch Time was set to 180 seconds to keep session duration consistent.

Procedure. Before the session, each participant received a brief standardized introduction to the study procedure and provided consent. Participants were asked to browse short-video content in a natural manner. They were allowed to end the session on their own. If a participant chose to stop before this limit, the session ended at that point; otherwise, the session ended when the maximum duration was reached.

4.3 Measurement

Three factors were measured immediately after the session: closure, perceived control, and affective appraisal. All Likert-style items were rated on 5-point scales

Closure was assessed using a single post-use Likert-type item. This approach follows Alexandrovsky et al who treated closure as a direct evaluative judgment [5].

Perceived control was measured using Perceived Choice subscale of the Intrinsic Motivation Inventory (IMI) [16].

Affective appraisal was measured by the Self-Assessment Manikin (SAM), using Valence and Arousal dimensions [17].

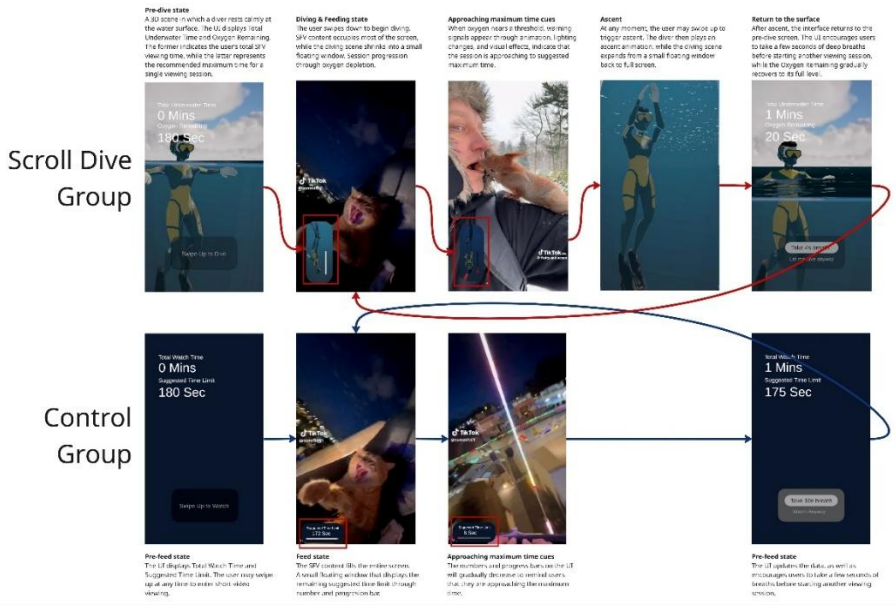


Fig. 1. Interaction flow in Scroll Dive and control condition.

4.4 Results

Perceived choice and arousal satisfied the parametric analysis and were therefore compared between conditions using independent-samples t-tests. Closure and valence did not meet the normality assumption, so they were analyzed using the Mann–Whitney U test.

Compared with the frictional design condition, Scroll Dive significantly increased arousal and produced a borderline significant increase in perceived choice, suggesting that the prototype made disengagement more activating and somewhat more autonomy supportive. However, no significant differences were found for closure or valence, which may partly reflect the limited sample size of the present study. Although both measures were directionally higher in the Scroll Dive condition, the present data do not support the claim that the prototype reliably improved the sense of completion or the pleasantness of exit.

Table 1. Results of independent-samples t-tests for perceived choice and arousal.

Measure	Scroll Dive Mean	Control group Mean	*t* statistics	*p*-value
Perceived choice	3.37(0.29)	3.16(0.15)	2.08	.050*
Arousal	3.83(0.58)	2.55(0.52)	5.59	.001*

Table 2. Results of Mann–Whitney U tests for closure and valence.

Measure	Scroll Dive Mean	Control group Mean	*Statistic*	*p*-value
Closure	Mean rank = 13.04	Mean rank = 10.86	$U = 53.50$, $Z = -0.85$	.394

Valence	Mean rank = 13.63	Mean rank = 10.23	$U = 46.50$, $Z = -1.56$	.119

5 Insights and Future Work

5.1 Preliminary insights

Overall, the findings suggest that Scroll Dive was more effective at making disengagement feel noticeable and agentic than the frictional control condition. These results offer several insights as follows.

The gameful metaphor may provide a make-believe frame in which stopping becomes a natural progression.

The prototype creates a make-believe frame in which oxygen depletion gives disengagement an internally coherent rationale. This makes stopping feel less like imposed persuasion and more like a natural next step within the experience.

Deliberate design for disengagement may make the stopping more noticeable rather than more pleasant.

The strongest effect of the prototype was found in arousal rather than valence or closure. This suggests that, in short-video contexts, designing for disengagement may initially work by making the ending of use more perceptually and experientially prominent, rather than by directly making it feel more positive or resolved.

Closure may be harder to support in short-video use than a well-structured ending alone can achieve.

The absence of a significant difference in closure suggests that even a meaningful and metaphorically organized disengagement process does not automatically produce a stronger sense of completion. Because short-video viewing is often continuous, open-ended, and lacks a clear endpoint, closure may be fundamentally challenging to design for in this context.

5.2 Future work

Future work should examine the prototype under more ecologically valid and longer-term conditions, including repeated use in contexts closer to everyday SFV browsing, to determine whether its observed effects remain stable beyond a single-session evaluation.

It would also be valuable to compare the diving metaphor with alternative disengagement metaphors or framing strategies, in order to clarify whether its effects are specific to this design or reflect a broader principle of metaphorically disengagement support. In addition, qualitative approaches could provide deeper insight into how users perceive the disengagement process and make sense of the metaphor.

Finally, future studies may expand the assessment of disengagement in SFV use beyond the current measures by considering additional factors, e.g., satisfaction with stopping, perceived interruption, post-exit reflection, etc.

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